SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: Total Marks: 10

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

```
A. <media src="video.mp4"></media>
```

```
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>
```

Snippet 2: Form Submission Methods

```
A. GET
B. POST
C. PUT
D. DELETE
```

Snippet 3: Navigation Elements

```
A. <div>
B. <nav>
C. <section>
D. <header>
```

Snippet 4: XML Syntax

```
A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note
```

Snippet 5: Web Communication Protocols

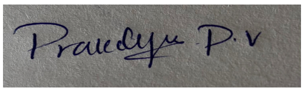
```
A. FTP
B. HTTP
C. SMTP
D. SNMP
```

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer. 2. Identify the **appropriate HTTP method for form submission** and explain why. 3. Choose the **proper semantic element for navigation** and justify its usage. 4. Identify the **correct XML syntax** and explain the error in other options. 5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

I P V Prakalya (192311418)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Scenario: Web Development Standards Implementation

1. Select the correct HTML tag for video embedding and justify your answer.

Correct Answer: B. `<video src="video.mp4" controls></video>`

Explanation:

The `<video>` tag is the standard HTML5 element used to display videos on a webpage. It supports different video formats such as MP4, WebM, and OGG. The `controls` attribute allows users to play, pause, adjust the volume, and view the video in full-screen mode.

Example:

```
<video src="video.mp4" controls width="600">
  Your browser does not support the video tag.
</video>
```

Why the other options are incorrect:

- **A. `<media>`** – This is not a valid HTML element.
- **C. `<movie>`** – HTML does not provide a `<movie>` tag.
- **D. `<vid>`** – This is also not a valid HTML tag.

Conclusion:

The `<video>` tag is the correct choice because it is officially supported in HTML5 and provides built-in controls for multimedia playback.

2. Identify the appropriate HTTP method for form submission and explain why.

Correct Answer: B. POST

Explanation:

The **POST** method is mainly used when users submit information through forms, such as registration, login, or payment forms. Unlike GET, the submitted data is placed inside the request body instead of the URL.

Advantages of POST:

- Protects sensitive information like passwords.
- Supports larger amounts of data.
- Data is not visible in the browser's address bar.
- Suitable for creating new records in a database.

Why the other options are incorrect:

- **A. GET** – Used mainly to retrieve information. Data appears in the URL.
- **C. PUT** – Used to update existing resources in REST APIs.
- **D. DELETE** – Used to remove resources from the server.

Conclusion:

For a registration form, POST is the most appropriate method because it provides better privacy and security.

3. Choose the proper semantic element for navigation and justify its usage.

Correct Answer: B. <nav>

Explanation:

The `<nav>` element is a semantic HTML5 tag specifically designed to contain navigation links. It clearly indicates that the enclosed links belong to the website's navigation section.

Example:

```
<nav>
  <a href="home.html">Home</a>
  <a href="about.html">About</a>
  <a href="services.html">Services</a>
  <a href="contact.html">Contact</a>
</nav>
```

Using semantic tags improves:

- Accessibility for screen readers.
- Search engine optimization (SEO).
- Code readability and maintenance.

Why the other options are incorrect:

- **A. `<div>`** – A generic container without semantic meaning.
- **C. `<section>`** – Used to group related content, not navigation.
- **D. `<header>`** – Represents introductory content and may contain navigation, but it is not specifically meant for navigation menus.

Conclusion:

The `<nav>` tag is the correct semantic element because it clearly identifies navigation links within a webpage.

4. Identify the correct XML syntax and explain the error in other options.

Correct Answer:

B.

```
<note>
  <to>User</to>
  <from>Admin</from>
</note>
```

Explanation:

XML requires every opening tag to have a matching closing tag, and all elements must be properly nested. Option B satisfies all XML syntax rules.

Errors in the other options:

Option A

```
<note><to>User</to><from>Admin</from>
```

Error: The closing `</note>` tag is missing.

Option C

<note><to>User</from></note>

Error: The opening tag is `<to>` but the closing tag is `</from>`. XML tags must match exactly.

<note to="User"></note

Errors:

- The closing angle bracket (`>`) is missing.
- The XML tag is incomplete.

Conclusion:

Option B is the only valid XML document because all tags are correctly opened, closed, and properly nested.

5. Choose the correct protocol for web communication and justify your choice.

Correct Answer: B. HTTP

Explanation:

HTTP (HyperText Transfer Protocol) is the standard communication protocol used between web browsers and web servers. It allows users to request webpages and receive responses from the server.

For secure communication, websites use HTTPS, which encrypts the transmitted data.

Why the other options are incorrect:

- **A. FTP** – Used for transferring files between computers.
- **C. SMTP** – Used for sending emails.
- **D. SNMP** – Used for monitoring and managing network devices.

Conclusion:

HTTP is the correct protocol because it is specifically designed for communication between web clients and web servers.

Overall Conclusion

In this case study, the startup should use modern web standards to build a reliable and secure application. The correct HTML5 `<video>` tag ensures proper multimedia support, while the **POST** method protects user information during form submission. Using the semantic `<nav>` element improves webpage structure and accessibility. Following proper XML syntax prevents parsing errors, and HTTP serves as the standard protocol for communication between browsers and web servers. By following these standards, the application becomes more secure, maintainable, and user-friendly.